TAO PeopleNet person object detection

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1. Case goal and processing flow
2. The checks before running it
3. Start the detection
4. The real-time detection with a USB camera
5. The real-time detection with a CSI camera
6. Frequently asked questions
7. References

> Applicable environment: Jetson Orin NX, JetPack 7.2.1, DeepStream 9.1.

1. Case goal and processing flow

PeopleNet detects the categories such as person, face and bag in the video frames, and outputs the category, the confidence and the rectangular box. The course configuration adds an NvDCF Tracker after the detection, so the text such as `person 3` in the frame carries the track ID inside the current video. It does not output a pixel-level human silhouette.

```
Video/USB/CSI → PeopleNet TensorRT inference → the OSD detection boxes
                                             ├→ the desktop preview
                                             └→ the sink output in the video mode
```

Use PeopleSemSegNet if you need the human silhouette; for the cross-camera appearance matching, keep using ReIdentificationNet. The Tracker ID is only valid inside the current video stream.

2. The checks before running it

The video mode runs in the graphical session of the Jetson desktop/VNC:

```
cd ~/deepstream_tao_apps/deepstream_app_tao_configs

# Confirm the configuration and the input source
grep -E "uri=|config-file=" run_peoplenet.txt

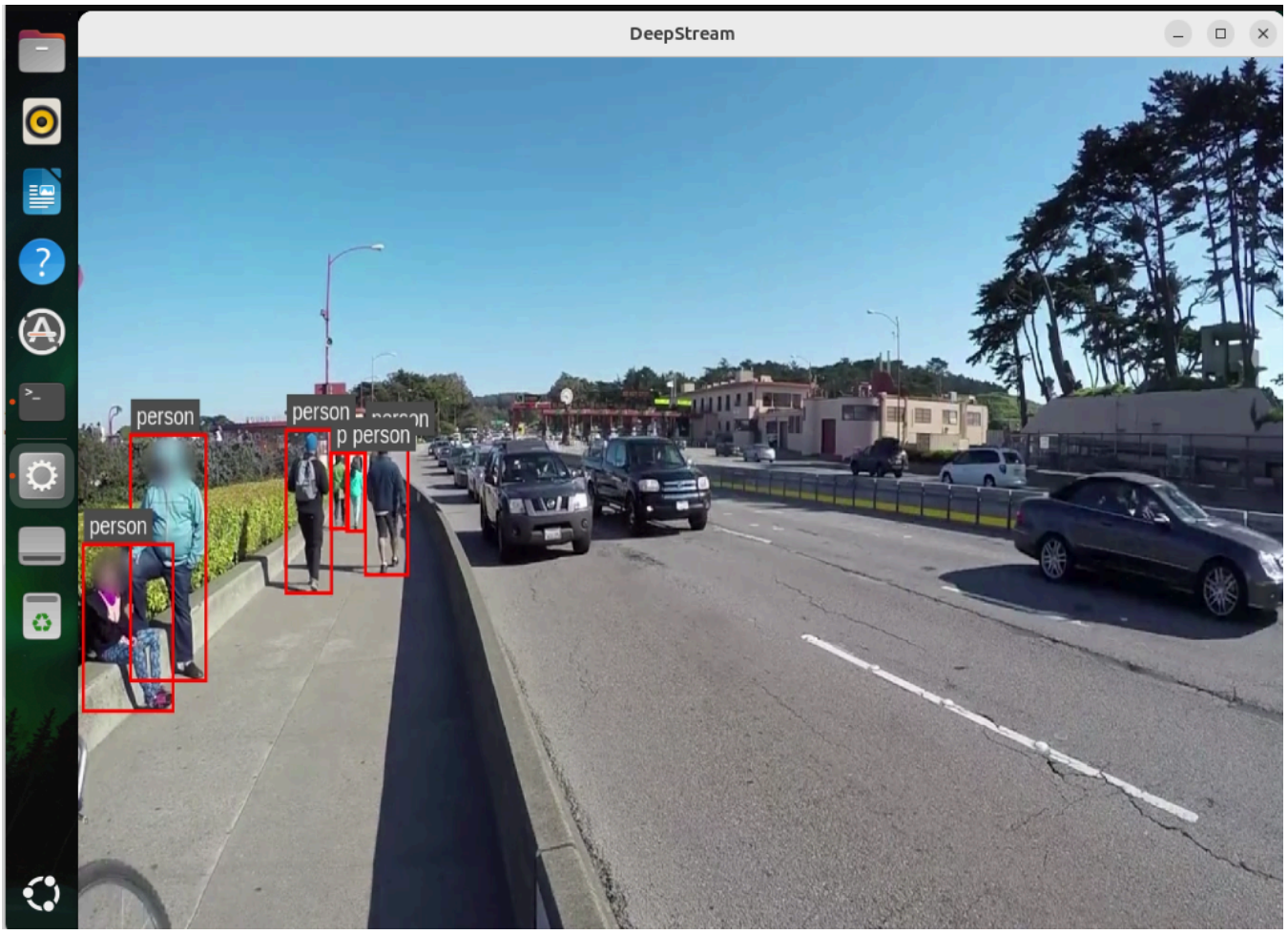
# Confirm that the model engine is valid
ls -la /opt/nvidia/deepstream/deepstream-
9.1/samples/models/tao_pretrained/peoplenet_vdeployable_quantized_onnx_v2.6.3/
```

Confirm that the material exists. The USB/CSI mode also needs the camera to be connected and the device to be confirmed as not being used by another program.

3. Start the detection

Run the built-in video:

```
export DISPLAY=:0 && deepstream-app -c
~/deepstream_tao_apps/deepstream_app_tao_configs/run_peoplenet.txt
```



The current video input is the sample_720p.h264 that comes with DeepStream. The detection boxes are displayed frame by frame; the category text depends on the labels file and on the order of the category IDs.

Wait for the EOS and you can see the result of the run; if the configuration has a file sink, the output goes to the corresponding path.

The input file is specified by the `uri=file://` of the `[source0]` section. When you replace the video, modify that URI directly to the absolute path of the MP4/H.264 file on the Jetson.

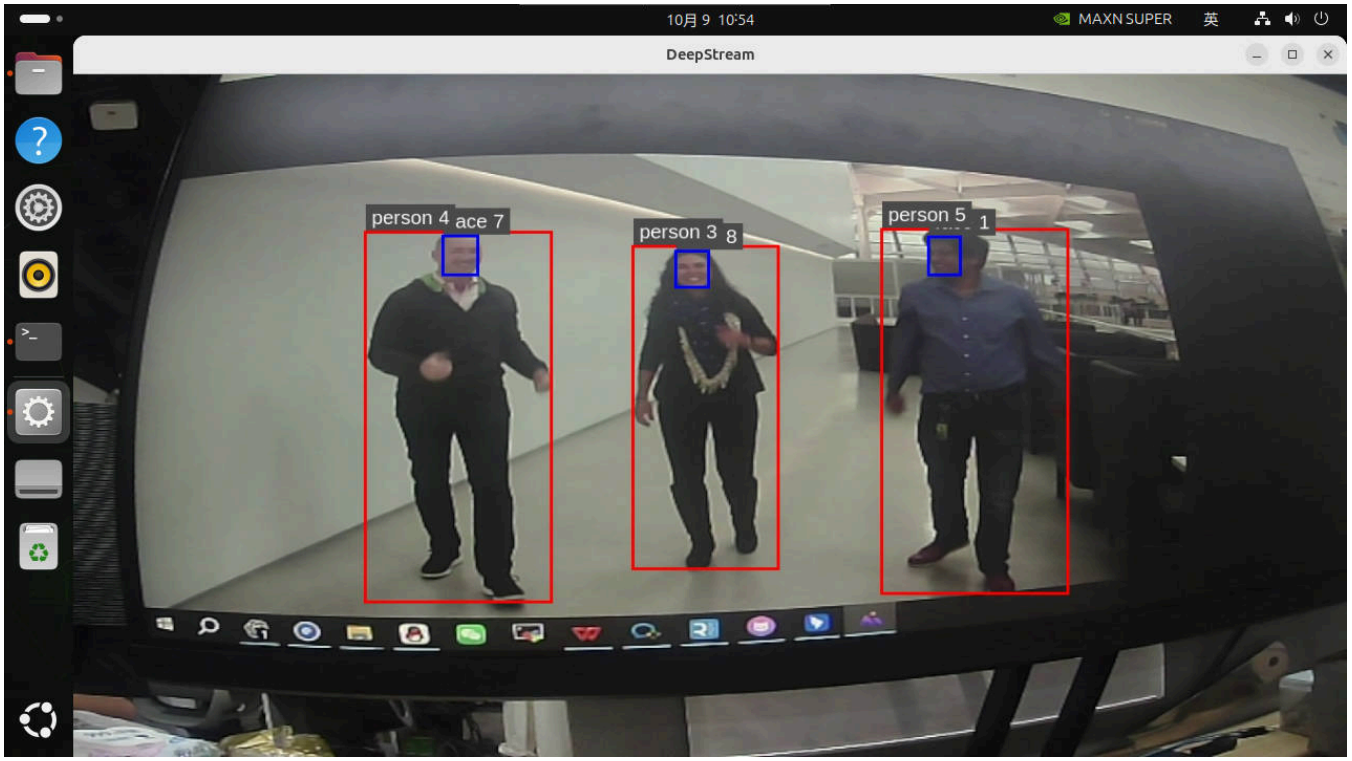
4. The real-time detection with a USB camera

The USB entry of the course supports PeopleNet. After connecting the camera you can check the nodes and the formats first:

```
ls -l /dev/video*
v4l2-ctl --list-devices
v4l2-ctl -d /dev/video0 --list-formats-ext
```

Start it from the Jetson desktop/VNC:

```
export DISPLAY=:0 && ~/deepstream_tao_demo/run_usb.sh peoplenet
```



The script detects the USB camera node automatically through `~/deepstream_tao_demo/find_usb_cam.sh`, and there is no need to specify it manually.

The USB mode is a real-time preview and it does not produce the MP4 of this case. The categories, the confidence, the boxes and the Tracker ID in the frame are the same as in the video mode; the actual node, the resolution and the frame rate are detected dynamically by the script.

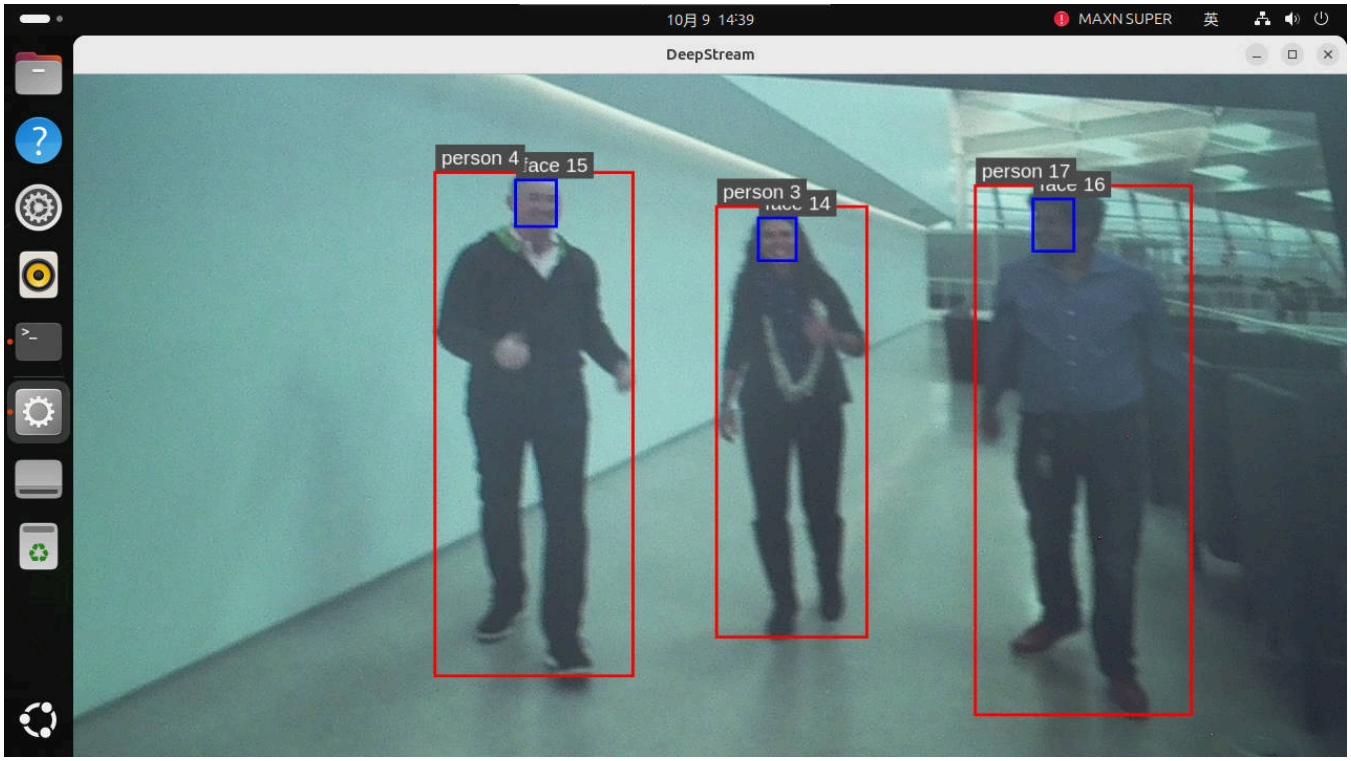
To stop it, press `Ctrl+C` once in the terminal that started it, wait for `App run successful` and confirm that the device has been released. The script prefers the `640x480@15` that the camera supports, and it automatically selects another common mode when it is not supported.

5. The real-time detection with a CSI camera

The CSI entry supports the real-time preview. After connecting a compatible camera and confirming that the driver and the Argus service work normally, run the following in a graphical terminal of the Jetson desktop/VNC:

Run:

```
~/deepstream_tao_demo/run_case.sh peoplenet csi
```



CSI uses the Jetson Argus link; the existence of a USB device node does not prove that the CSI streaming works normally. The camera mode is used for the real-time preview.

6. Frequently asked questions

Symptom	The check and the handling
There are no person boxes	Confirm that the input frame has a target and that the pipeline is running; the occlusion, the scale and the lighting all affect the detection.
The categories are displayed wrongly	Check whether the PeopleNet engine, the inference configuration, the parser and the labels all come from the same version.
The USB cannot be opened	Check `/dev/video*`, the supported formats, the permissions and whether the device is in use.
There is no image with CSI	Check the compatible module, the connection, the sensor driver, the Argus service and whether the camera is in use. The script releases the pose/OCR CSI relay automatically.
The MP4 is missing or cannot be seeked	The camera mode only previews; in the video mode you have to wait for the EOS or check after a single Ctrl+C has finished it.
`nvv4l2h264enc has no property named 'gpu-id'`	It is a known non-blocking warning of DS 9.1; it appears when the OSD sink is initialized and it does not affect the inference or the display.

7. References

- The NVIDIA DeepStream TAO application examples: https://github.com/NVIDIA/DeepStream/tree/main/src/apps/tao_apps
- The DeepStream TAO integration documentation: https://docs.nvidia.com/metropolis/deepstream/dev-

guide/text/DS_TAO_integration.html